Apalutamide Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Apalutamide is indicated for the treatment of adult men with non-metastatic castration resistant prostate cancer (nmCRPC) who are at high risk of developing metastatic disease.	C61	00574a	CDS 01/08/2021
For the treatment of patients with metastatic hormone sensitive prostate cancer (mHSPC) in combination with androgen deprivation therapy.	C61	00574b	CDS 01/04/2023

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patient's individual clinical circumstances.

Apalutamide is to be taken orally once daily.

Cycle frequency and duration:

Treatment is continuous until disease progression or unacceptable toxicity.

Drug	Dose	Route	Cycle
Apalutamide[a]	240mg once daily[b]	PO with or without food at the same time each day.[c] Tablets should be swallowed whole with water.[d]	Continuous
[a]Medical castration with gonadotropin releasing hormone analogue (GnRHa) should be continued during treatment in patients not surgically castrated.			
[b]Apalutamide is commonly available as 240mg and 60mg film-coated tablets.			
[c]If a dose is missed, it should be taken as soon as possible on the same day with a return to the normal schedule the following day. Extra tablets should not be taken to make up the missed dose.			
[d]Please refer to SmPC for advice on administration of apalutamide to patients who cannot swallow the tablet whole, and for advice on administration via nasogastric feeding tube.			

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ELIGIBILITY:

- Indications as above
- ECOG 0-2
- Adequate haematologic, hepatic, and renal function
- **nmCRPC:**
    - Histologically or cytologically confirmed adenocarcinoma of the prostate without neuroendocrine differentiation or small cell features with high risk for development of metastases, defined as prostate-specific antigen doubling time (PSADT) ≤ 10 months
    - Castration-resistant prostate cancer demonstrated during continuous ADT, defined as 3 PSA rises, at least 1 week apart, with the last PSA greater than > 2ng/mL
    - Surgically or medically castrated, with testosterone levels of < 50ng/dL

- **mHSPC:**
    - Histologically or cytologically confirmed adenocarcinoma of the prostate, and distant metastatic disease consisting of at least one lesion on bone scanning (with or without visceral or lymph node involvement)
    - Patients must be castration sensitive (i.e. not receiving ADT at the time of disease progression)

EXCLUSIONS:

- Hypersensitivity to apalutamide or any of the excipients
- Prior treatment with second generation anti-androgens
- Prior treatment with CYP17 inhibitors
- History of seizure or condition that may pre-dispose to seizure
- Clinically significant cardiovascular disease
- **nmCRPC**
    - Prior chemotherapy for prostate cancer except if administered in the adjuvant/neoadjuvant setting

PRESCRIPTIVE AUTHORITY:

- The treatment plan must be initiated by a Consultant Medical Oncologist

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TESTS:

Baseline tests:

- FBC, renal and liver profile
- Thyroid function
- ECG in patients at risk of QT prolongation
- INR for patients on warfarin

Regular tests:

- FBC, renal and liver profile
- ECG as clinically indicated
- Weekly INR tests if patient is on warfarin until stable warfarin dose established
- Thyroid function as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant.

Table 1: Dose modification of apalutamide

Dose Level	Recommended Dose Modification
0	240mg
-1	180mg
-2	120mg

Table 2: Dose modification of apalutamide in renal and hepatic impairment

Renal Impairment		Hepatic Impairment	
CrCl (mL/minute)	**Dose**	**Child-Pugh Class**	**Dose**
≥ 30	No dose adjustment is needed	A/B	No dose adjustment is necessary
< 30	No need for dose adjustment is expected	C	120mg daily is recommended
Haemodialysis	No need for dose adjustment is expected		
Renal dose modification recommendations obtained from Giraud et al. 2023. Hepatic dose modification recommendations obtained from SmPC.			

Table 3: Dose modification of apalutamide for adverse events

Adverse reactions	Recommended dose modification
Intolerable or ≥ Grade 3	Withhold until symptoms improve to ≤ Grade 1 or original grade, then treatment may be resumed at the same dose or a reduced dose (180mg or 120mg), if warranted.
Seizures	Discontinue
Interstitial lung disease	Discontinue

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SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Apalutamide: Minimal-low **(Refer to local policy).**

For information:
Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS:

None usually required

OTHER SUPPORTIVE CARE:

Apalutamide has no or negligible influence on the ability to drive and use machines. However, seizures have been reported in patients taking apalutamide. Patients should be advised of this risk in regards to driving or operating machines.

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for full details of adverse effects.
- Please note the NCCP National SACT Regimens do not provide information regarding black triangle (▼) status. Please refer to the relevant Summary of Product Characteristics (SmPC) for this information.

REGIMEN SPECIFIC COMPLICATIONS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details
- **Seizure**: Apalutamide is not recommended in patients with a history of seizures or other predisposing factors including, but not limited to, underlying brain injury, recent stroke (within one year), primary brain tumours or brain metastases. If a seizure develops during treatment with apalutamide, treatment should be discontinued permanently. The risk of seizure may be increased in patients receiving concomitant medicinal products that lower the seizure threshold. Seizure occurred in 0.2% of patients receiving apalutamide in clinical studies. These studies excluded patients with a history of seizure or predisposing factors for seizure. There is no clinical experience in re-administering apalutamide to patients who experienced a seizure.
- **Concomitant use with other medicinal products:** Apalutamide is a potent enzyme inducer and may lead to loss of efficacy of many commonly used medicinal products. A review of concomitant medicinal products should therefore be conducted when apalutamide treatment is initiated. Concomitant use of apalutamide with medicinal products that are sensitive substrates of many metabolising enzymes or transporters should generally be avoided if their therapeutic effect is of large importance to the patient, and if dose adjustments cannot easily be performed based on monitoring of efficacy or plasma concentrations. Co-administration with warfarin and coumarin-like anticoagulants should be avoided. If apalutamide is co-

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administered with an anticoagulant metabolised by CYP2C9 (such as warfarin or acenocoumarol), additional International Normalised Ratio (INR) monitoring should be conducted.

- **Recent cardiovascular disease:** Patients with clinically significant cardiovascular disease in the past 6 months including severe/unstable angina, myocardial infarction, symptomatic congestive heart failure, arterial or venous thromboembolic events (e.g. pulmonary embolism, cerebrovascular accident including transient ischemic attacks), or clinically significant ventricular arrhythmias were excluded from the clinical studies. Therefore, the safety of apalutamide in these patients has not been established. If apalutamide is prescribed, patients with clinically significant cardiovascular disease should be monitored for risk factors such as hypercholesterolaemia, hypertriglyceridaemia, or other cardio-metabolic disorders. Patients should be treated, if appropriate, after initiating apalutamide for these conditions according to established treatment guidelines.
- **Androgen deprivation therapy may prolong the QT interval:** In patients with a history of or risk factors for QT prolongation and in patients receiving concomitant medicinal products that might prolong the QT interval, physicians should assess the benefit-risk ratio including the potential for Torsade de pointes prior to initiating apalutamide.
- **Hypothyroidism:** Hypothyroidism was reported for 8% of patients treated with apalutamide and 2.0% of patients treated with placebo based on assessments of thyroid-stimulating hormone (TSH) every 4 months. There were no grade 3 or 4 adverse events. Hypothyroidism occurred in 30% of patients already receiving thyroid replacement therapy in the apalutamide arm and in 3% of patients in the placebo arm. In patients not receiving thyroid replacement therapy, hypothyroidism occurred in 7% of patients treated with apalutamide and in 2% of patients treated with placebo. Thyroid replacement therapy, when clinically indicated, should be initiated or dose-adjusted.
- **Skin rash:** Skin rash associated with apalutamide was most commonly described as macular or maculo-papular. Skin rash included rash, rash maculo-papular, rash generalised, urticaria, rash pruritic, rash macular, conjunctivitis, erythema multiforme, rash papular, skin exfoliation, genital rash, rash erythematous, stomatitis, drug eruption, mouth ulceration, rash pustular, blister, papule, pemphigoid, skin erosion, and rash vesicular. Adverse reactions of skin rash were reported for 26% of patients treated with apalutamide. Grade 3 skin rashes (defined as covering > 30% body surface area [BSA]) were reported with apalutamide treatment in 6% of patients.
- **Stevens-Johnson syndrome (SJS)/toxic epidermal necrolysis (TEN):** Postmarketing reports of SJS/TEN, which can be life-threatening or fatal, have been observed in association with apalutamide treatment. Patients should be advised of signs and symptoms suggestive of SJS/TEN. If these symptoms are observed, apalutamide should be withdrawn immediately and patients should seek immediate medical consultation. Apalutamide must not be restarted in patients who have experienced SJS/TEN while taking apalutamide at any time and an alternative treatment should be considered.
- **Interstitial lung disease (ILD):** Cases of ILD have been observed in patients treated with apalutamide, including fatal cases. In case of acute onset and/or unexplained worsening of pulmonary symptoms, treatment with apalutamide should be interrupted pending further investigation of these symptoms. If ILD is diagnosed, apalutamide should be discontinued and appropriate treatment initiated as necessary.

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DRUG INTERACTIONS:

- Current drug interaction databases and SmPC should be consulted for more information.

REFERENCES:

1. Saad F. Effect of apalutamide on health-related quality of life in patients with non-metastatic castration-resistant prostate cancer: an analysis of the SPARTAN randomised, placebo-controlled, phase 3 trial. Lancet Oncol. 2018 Oct;19(10):1404-1416. https://doi.org/10.1016/S1470-2045(18)30456-X. Including supplementary appendix and protocol.
2. Kim N, et al. Apalutamide for Metastatic, Castration-Sensitive Prostate Cancer. N Engl J Med 2019; 381:13-24. Available at: https://www.nejm.org/doi/full/10.1056/nejmoa1903307
3. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
4. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). Antiemesis. Version 2.2025 – May 12, 2025. Accessed 24/12/2025. Available at: https://www.nccn.org/guidelines/guidelines-detail?category=1&id=1415
5. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
6. Apalutamide (Erleada®) 60mg film-coated tablets. SmPC EMA. Last updated: 22/04/2026. Accessed 29/05/2025. Available at: https://www.ema.europa.eu/en/documents/product-information/erleada-epar-product-information_en.pdf

From 01/07/2026, this regimen will be reviewed on an 'As Required' basis. Please refer to the version control box for information, and the footer for date of last review. If you wish to suggest any updates to this regimen, please contact oncologydrugs@cancercontrol.ie

Version	Date	Amendment	Approved By
1	08/01/2020		Prof. Maccon Keane
2	06/01/2021	Reviewed	Prof. Maccon Keane
3	01/08/2021	Reimbursement status updated	NCCP
4	01/04/2023	Added new indication. Updated adverse events section	Prof. Maccon Keane
4a	19/3/2025	Amended exclusions section to clarify nmCRPC related exclusions	NCCP
5	01/07/2026	Updated regimen review schedule to 'as required'. Added interstitial lung disease to regimen specific complications and dose modification section (Table 3). Removed falls and fractures from regimen specific complications. General NCCP Standardisation updates.	Prof. Maccon Keane

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		Added footnotes to treatment table regarding missed doses and administration via nasogastric tube. Updated ECOG to 0-2, exclusions section, updated renal and hepatic impairment section (renal as per Giraud et al 2023, hepatic as per SmPC.	

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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